DEPARTMENT OF STATISTICS
Diponegoro University, Indonesia

PRESENTED BY
NGANTUUUUUKKKKKKKKKKK TEAM

EVALUATING GLOBAL STUNTING PERFORMANCE

Applying a Bivariate Risk Compass Model for Prioritizing Country Interventions

"Data gives us insight, collaboration gives us strength, and our children's smiles are the ultimate measure of our success."

2 ZERO HUNGER

3 GOOD HEALTH AND WELL-BEING

6 CLEAN WATER AND SANITATION

The Global Stunting Crisis

The problem of stunting chronic malnutrition leading to impaired growth remains one of the most significant yet solvable global public health crises today. It is **directly tied to SDG 2 (Zero Hunger**), specifically Target 2.2, which calls for ending all forms of malnutrition by 2030. **However, progress remains dangerously slow,** threatening the future of human capital.

148 Million

Children under the age of five worldwide are stunted

As an irreversible condition, stunting severely restricts physical growth while more critically resulting in impaired cognitive development and reduced adult productivity *(WHO/UNICEF/World Bank Joint Malnutrition Estimates, 2023)*.

The Interconnected Burden: Economic and Multisectoral Dimensions

The issue is intrinsically **multisectoral**, necessitating analysis that incorporates **SDG 6 (Clean Water and Sanitation) and SDG 3 (Good Health)**, as environmental factors significantly drive nutritional outcomes. (World Bank Report on the Cost of Malnutrition).

The Need for a New Analytical Framework

Traditional policy ranking often relies on simple prevalence data, which provides a snapshot of the Outcome but fails to inform strategic action regarding the Determinants of risk. By utilizing the **Weighted Sum Model (WSM)** to create a comprehensive Bivariate Risk Compass, we categorize countries based on their Outcome Index (Y) versus their Determinant Index (X).

Methodology: From Data to Actionable Risk

Robust scaling → KNN imputation → Indexing (WSM) → Weighting → α-blend → Compass → Clustering

Y-Axis (Outcome Index)

- **SH.STA.STNT.ZS (Stunting Prevalence)**
- **SH.STA.WAST.ZS (Wasting Prevalence)**

both scaled to 0-1

Why: direct child undernutrition outcomes required by SDG 2.2; captures current burden

X-Axis (Determinants Index)

- **SI.POV.DDAY (Extreme Poverty)**
- **SH.H2O.BASW.ZS (Access to Basic Water Services)**
- **SH.STA.BASS.ZS (Access to Basic Sanitation Services)**
- **SH.IMM.MCV1.ZS (Measles Immunization)**

all scaled to 0–1 with polarity correction (good-high inverted)

Why: poverty and WASH/health are the most actionable structural levers for reducing stunting

$$Index = \sum w_j \cdot indicator_j$$

All indicators are first put on a 0–1 risk scale (higher = worse)
Weights w are non-negative and sum to 1.

WORKFLOW

1) Normalize all indicators to a 0–1 risk scale and align polarity so “higher = more risk”

2) Impute missing values in the normalized space with KNN to retain countries without distorting distances

3) Construct two composite indices with WSM: Y (outcomes) and X (determinants) using non-negative weights that sum to 1

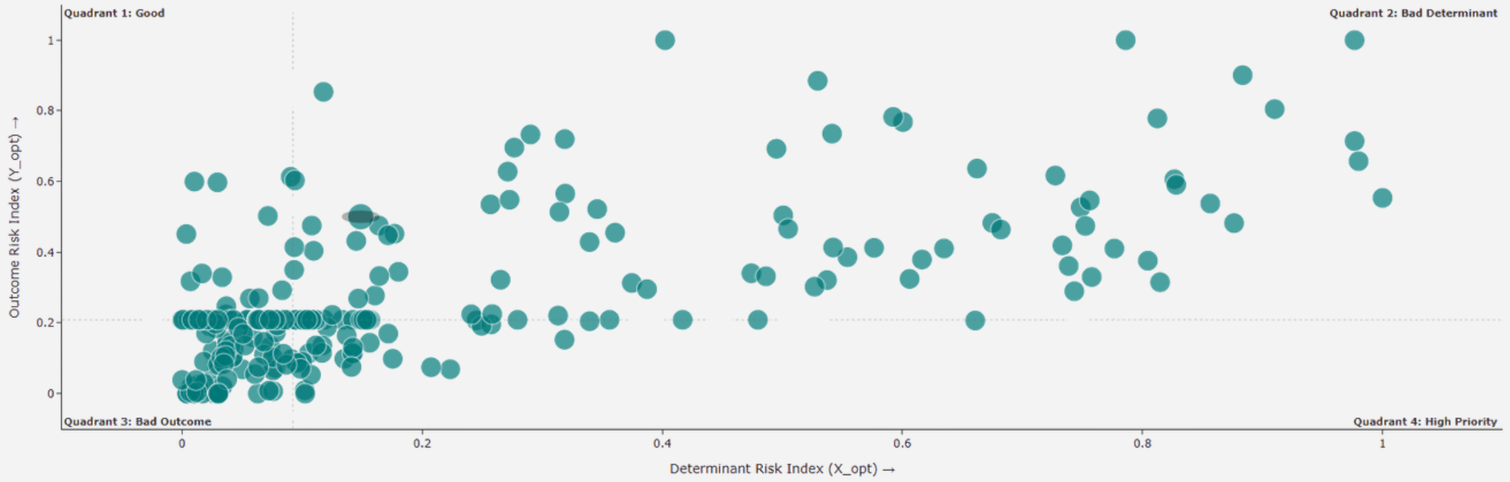
4) Evaluate expert/entropy/PCA/regression weights; keep the scheme that maximizes monotonicity with Y and cross-country separation. blacky

5) Compute with $R(\alpha)=\alpha Y+(1-\alpha)$ and select α for strongest behavior (monotonicity and separation)

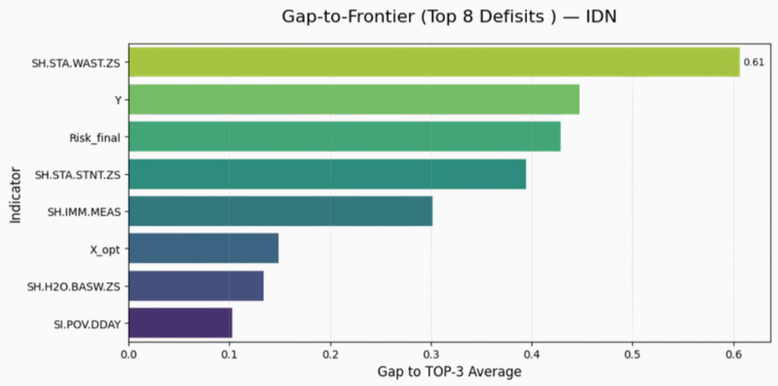
6) Visualize the Risk Compass (X vs Y) with dashed medians and highlight the focus country (Indonesia) to read priorities at a glance

7) Group countries into three levels (Low/Medium/High) by applying 1-D GMM with Bayesian Information Criterion check

Stunting Risk Compass Global (264 Country)

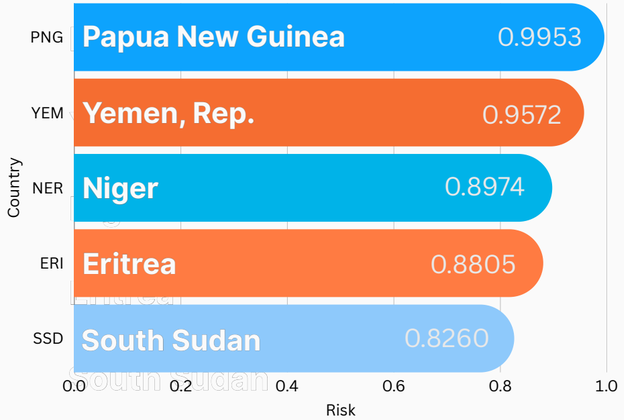


How About Indonesia?



Benchmarking Global Risk (WSM Output)

TOP 5 Priority Zones by Final Risk Index



Country Code	Country Name	Risk_final	X_opt	Y_opt
KOR	Korea, Rep.	0.0008	0.0042	0.0000
DEU	Germany	0.0008	0.0042	0.0000
BEL	Belgium	0.0013	0.0067	0.0000
MLT	Malta	0.0020	0.0100	0.0000

Countries at Near-Zero Risk

Korea and Germany are the global benchmark. They have achieved near-zero stunting risk because their health and structural systems are exceptionally well managed.

Program Inefficiency and the Wasting Crisis

The data reveals a critical issue of program inefficiency

Y (Outcome) = 3 Times X (Determinan Gap)

This disparity suggests that **although Indonesia is successfully reducing structural risk factors** (e.g., poverty, sanitation), **the conversion of these improvements into actual nutritional outcomes remains highly inefficient.**

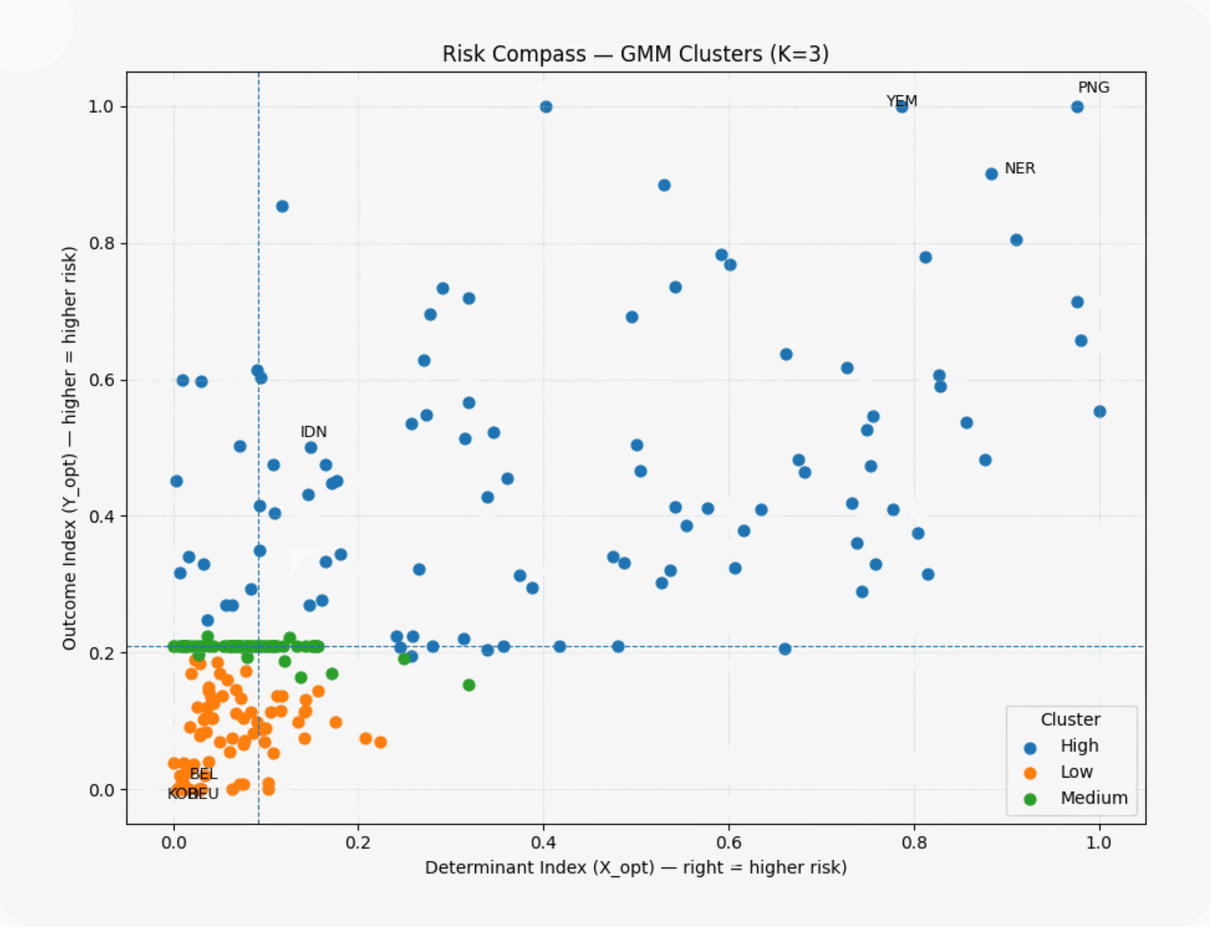
Clustered Risk Profiles of Countries

This Risk Compass visualization applies a **Gaussian Mixture Model (GMM)** with K = 3 clusters to classify countries based on their overall stunting risk.

Cluster:

- **Low-risk cluster:** Countries with strong determinants and low stunting prevalence.
- **Medium-risk cluster:** Countries with moderate determinants and outcomes.
- **High-risk cluster:** Countries with both weak determinants and severe stunting outcomes.

Indonesia falls within the **High-Risk cluster**, meaning that despite ongoing improvements, the response systems (nutrition, WASH, health access) remain insufficient to match the burden.



Conclusion

Stunting risk is not defined solely by prevalence but by a country's capacity to respond. Strengthening determinants like poverty reduction, water, sanitation, and health systems is key to achieving lasting progress toward ending stunting.

"Risk is not only about how many are affected, but how ready we are to respond."

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